
Acceptance sampling

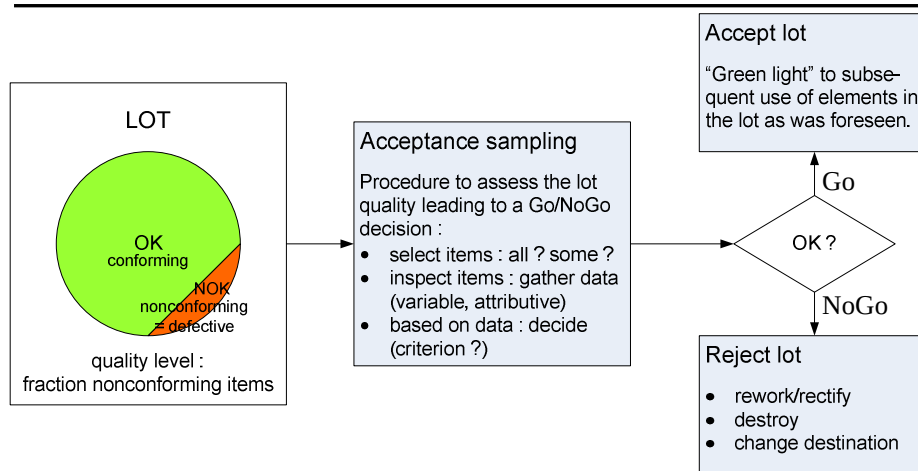
Sampling plans for product inspection

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1

The acceptance sampling procedure



AQL : acceptable quality level – should normally pass the inspection

LQL : limiting quality level (LTPD : lot tolerance percent defectives) – should normally be rejected

Producer's risk : risk of rejecting a good lot

Consumer's risk : risk of accepting a good lot

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2

On the quality ladder sampling plans are situated on the 2nd level, which is the lowest level of systematic quality assurance methods. Sampling plans are important in those situations where products are received from suppliers (either external or internal) on whom there is no other assessment of the quality level of their production processes. When suppliers have established procedures of statistical process control, then (sampling) inspection of shipments may not be necessary. In such case a supplier will demonstrate that his production process maintains a capability level in accordance with the customer's requirements and as a consequence its products are delivered directly, without inspection, to the production line. But even for certified suppliers it may be a prudent strategy to periodically audit the quality level to ensure that this doesn't drop below acceptable standards (supplier evaluation).

A lot is a collection of N elements, i.e. a finite real population of products. The quality level of the lot can be described by the proportion of nonconforming (synonym : defective) items in the lot.

A decision needs to be made as to whether a lot has to be subjected to complete inspection (this is often called screening) or whether acceptance can be determined using only a sample from the lot. When using a sample, the sampling plan, the size and the criterion for acceptance need to be determined.

The criterion for acceptance of a lot will either be variable or attributive. In the latter case each item after inspection will be classified as either OK or NOK. The test conditions to classify the items have to be well determined, i.e. an operational definition of quality has to be available. Such an operational definition specifies in detail how a quality characteristic will be measured.

100 % inspection will be necessary when items are used in critical or very expensive systems, i.e. when the cost of inspection is negligible compared to the expected cost of failure. In many situations 100 % inspection is not the ideal solution : it may not be possible (destructive testing) or not practical (expensive). In this case inspection will be based on a sample drawn from the lot. In this situation the decision is taken without knowing the exact quality level in the lot. The procedure should satisfy the requirement that lots with acceptable quality have a good chance to pass the inspection, while lots with unacceptable quality have little chance to pass the inspection. Acceptable quality is described by the number AQL (acceptable quality level), while unacceptable quality is described by the number LQL (limiting quality level, synonym : LTPD = lot tolerance percent defectives). AQL and LQL should be agreed upon by both the parties involved.

The risks involved in the procedure are called : producer's risk (risk of rejecting a lot of good quality = type-I error) and consumer's risk (risk of accepting a lot of bad quality = type-II error).

Contents

- ⇒ Single-stage sampling plans for attributes
- ⇒ Double sampling plans for attributes
- ⇒ Sequential sampling for attributes
- ⇒ Sampling plans for variables
- ⇒ Rectifying inspection
- ⇒ Skip-lot sampling plans for attributes
- ⇒ International standards – sampling systems
- ⇒ Deming's inspection criterion
- ⇒ Certified quality provider
- ⇒ Exercises

Single-stage sampling plans for attributes

- ⇒ Take 1 RSWOR of size n , determine the number X of defective items, accept the lot if $X \leq c$. (c = acceptance number). Notation (n, c) .
- ⇒ $OC(p) = P(X \leq c | p) = H(c; N, M_p, n)$ with $M_p = [Np]$.
- ⇒ AQL = p such that $OC(p) = 1 - \alpha$ (= 95 %)
- ⇒ LQL = p such that $OC(p) = \beta$ (= 10 %)
- ⇒ Specifying both AQL and LQL characterizes the plan.
 - approximate method K&Z § 9.3 p. 297
 - approximate method by Peach

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4

H = the hypergeometric c.d.f. To calculate the OC-curve in R, use `phyper(c, M, N-M, n)`.

The AQL of a sampling plan can be defined as the percentage defectives for which the plan gives an acceptance probability of $1 - \alpha$, normally $\alpha = 5\%$. In the same way for the LQL (LTPD), normally taken $\beta = 10\%$.

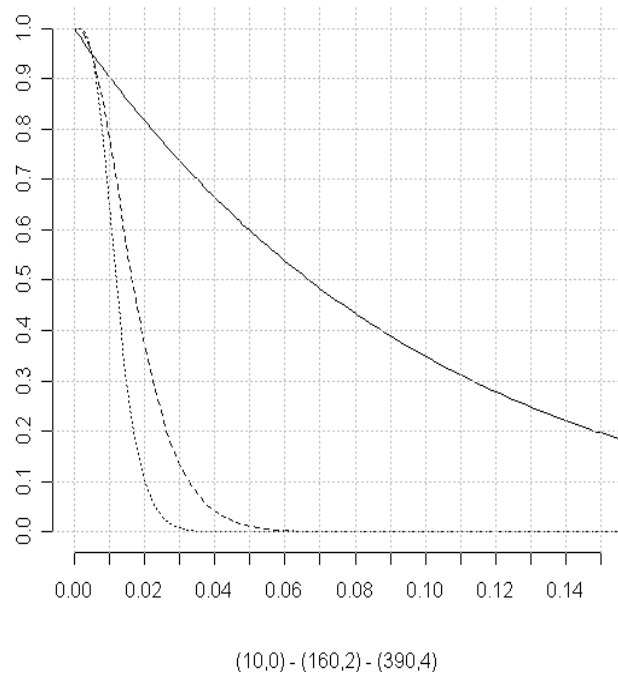
Specifying both AQL and LQL characterizes the plan, i.e. there is only one (n, c) with given AQL and LQL. For the determination of the plan, there are several methods. Here we describe the method of Peach (see also "tabellen AS.pdf").

This method can find single plans for which the OC-curve goes through $(p_1, 0.95)$ and $(p_2, 0.05)$. Form $R_0 = p_2/p_1$ and look for R_0 in the table. When the value cannot be found, look for the next larger tabled value.

Example : what plan satisfies (0.5 %, 95 %) and (3.5 %, 5 %) ? Here $R_0 = 7$. The tabled value is 7.7. So $c = 2$ and $n = 164$. This result is approximately confirmed by the chart on the right (plan (160,2)).

Notice that the OC-curve of a sampling plan satisfies the classical interpretation, it shows the risk of committing the type-II error, when regarding the inspection as a hypothesis test.

OC-curves for (n, c)



Double sampling plans for attributes

⇒ Take a RSWOR of size n_1 , determine the number X_1 of defective items, accept the lot if $X_1 \leq c_1$, reject the lot if $X_1 \geq c_2$. If X_1 is between c_1 and c_2 , take another RSWOR of size n_2 , determine the number X_2 of defective items. If $X_1 + X_2 \leq c_3$, accept the lot, else reject the lot. Notation $(n_1, c_1, c_2) + (n_2, c_3)$.

$$\Rightarrow OC(p) = H(c_1; N, M_p, n_1) + \sum_{j=c_1+1}^{c_2-1} h(j; N, M_p, n_1) H(c_3 - j; N - n_1, M_p - j, n_2)$$

$$\Pi(p) = H(c_1; N, M_p, n_1) + 1 - H(c_2 - 1; N, M_p, n_1)$$

$$ASN(p) = n_1 \Pi(p) + (n_1 + n_2)(1 - \Pi(p))$$

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5

In a single sampling plan the procedure requires one sample. In a double plan, a second sample may be required. But when the fraction defectives in the lot either is low or high, we may expect that the decision to accept or reject the lot will be reached after the first stage of sampling. Therefore double sampling plans are more efficient than single plans. This can be quantified by calculating the ASN-function. ASN = average sample number, it is the expected sample size for a given fraction defectives p .

The method of Peach can also be applied to determine a double plan (take $n_1 = n_2$). As an exercise : determine with this method a double plan with an OC-curve passing through (0.5 %, 95 %) and (3.5 %, 5%). Compare the OC-curve of this plan to the OC-curve of the plan (164,2). Next calculate the ASN-function of the plan and compare it to 164. What conclusions can be drawn ?

From the point of view of efficiency, double plans are better than single, but double plans take more effort to implement. In certain situations it may not be practically possible to organize a double plan.

Sequential sampling for attributes

Wald sequential probability ratio test (SPRT)

(For binomial distribution instead of hypergeometric, OK if lot size is large compared to sample size.)

$$\Lambda(X_n, n; p_0, p_t) = \left(\frac{p_t}{p_0} \right)^{X_n} \left(\frac{1-p_t}{1-p_0} \right)^{n-X_n} \quad \Lambda \text{ is the so-called likelihood ratio}$$

$$\log \Lambda = X_n \underbrace{\left(\log \left(\frac{p_t}{p_0} \right) \right)}_{>0} - \underbrace{\log \left(\frac{1-p_t}{1-p_0} \right)}_{<0} + n \log \left(\frac{1-p_t}{1-p_0} \right)$$

Behaviour of $\log \Lambda$ (see R-function SPRT.TXT) :

- starts at value 0 for $n = X_n = 0$
- X_n is determined by the true fraction defectives p
- $n: +1$ and $X_n: +0 \Rightarrow \log \Lambda \downarrow$ (small step)
- $n: +1$ and $X_n: +1 \Rightarrow \log \Lambda \uparrow$ (larger step)
- if p is large enough : the overall trend will be positive
- if p is small enough : the overall trend will be negative

stop procedure and decide :

if $\log \Lambda < \log A$ (for some $A < 1$) : accept
if $\log \Lambda > \log B$ (for some $B > 1$) : reject

$$\left. \begin{array}{l} A = \frac{\beta}{1-\alpha} < 1 \\ B = \frac{1-\beta}{\alpha} > 1 \end{array} \right\} \rightarrow \begin{array}{l} PR = 1 - OC(p_0) = \alpha \\ CR = OC(p_t) = \beta \end{array}$$

SPRT has the smallest ASN values at p_0 and p_t (optimality of the SPRT)

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6

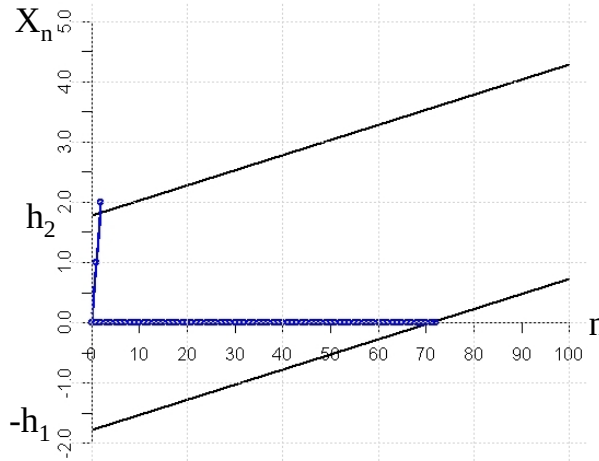
We now generalize the idea of double sampling to try to reach decisions with smaller inspection effort, and therefore at reduced cost. Sometimes it is possible to subject a selected item to immediate testing before selecting the next item. In this situation we can decide after each observation if sufficient information is available for accepting or rejecting the lot, or if an additional item should be randomly selected and tested. This approach may lead to substantial reduction in the number of required observations, especially when the proportion of nonconforming items in the lot is either very small or very high.

$C_n^{X_n} p^{X_n} (1-p)^{n-X_n}$ as a function of p , is the likelihood of p , given a number X_n of defectives in a sample of size n , so Λ is called the likelihood ratio.

Normally p_0 and p_t are small fractions, therefore in the expression of $\log \Lambda$, the coefficient of X_n is positive, while that one of n is negative ; in absolute value the coefficient of X_n is (much) bigger than that one of n . As n increases, X_n will on average increase too, according to $E(X_n) = np$. As the procedure goes on, each time n increases with one unit while X_n stays the same, $\log \Lambda$ decreases somewhat, but if X_n increases with one unit too (a defective item found), $\log \Lambda$ will show an "upward jump". Only if there are enough jumps to compensate for the gradual downward trend, $\log \Lambda$ will be on average increasing. Then the procedure is stopped and the lot is rejected when some level $\log B$ is attained. If the global trend is decreasing the procedure is stopped when $\log \Lambda$ goes below $\log A$: the lot is accepted.

Sequential sampling for attributes

SPRT : graphical representation



shortest path to accept :

$$-h_1 + sn = 0 \Leftrightarrow n = [h_1/s] + 1$$

shortest path to reject :

$$h_2 + sn = n \Leftrightarrow n = [h_2/(1-s)] + 1$$

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7

Reject :

$$\left(\log \left(\frac{p_t}{p_0} \right) - \log \left(\frac{1-p_t}{1-p_0} \right) \right) X_n \geq \log \left(\frac{1-\beta}{\alpha} \right) - n \log \left(\frac{1-p_t}{1-p_0} \right)$$

$$\Leftrightarrow \log \left(\frac{p_t(1-p_0)}{p_0(1-p_t)} \right) X_n \geq \log \left(\frac{1-\beta}{\alpha} \right) + n \log \left(\frac{1-p_0}{1-p_t} \right)$$

$$\Leftrightarrow X_n \geq h_2 + sn$$

$$\text{with } h_2 = \log \left(\frac{1-\beta}{\alpha} \right) / \log \left(\frac{p_t(1-p_0)}{p_0(1-p_t)} \right) \text{ and } s = \log \left(\frac{1-p_0}{1-p_t} \right) / \log \left(\frac{p_t(1-p_0)}{p_0(1-p_t)} \right) \text{ both } > 0$$

Accept :

$$\left(\log \left(\frac{p_t}{p_0} \right) - \log \left(\frac{1-p_t}{1-p_0} \right) \right) X_n \leq \log \left(\frac{\beta}{1-\alpha} \right) - n \log \left(\frac{1-p_t}{1-p_0} \right)$$

$$\Leftrightarrow \dots$$

$$\Leftrightarrow X_n \leq -h_1 + sn$$

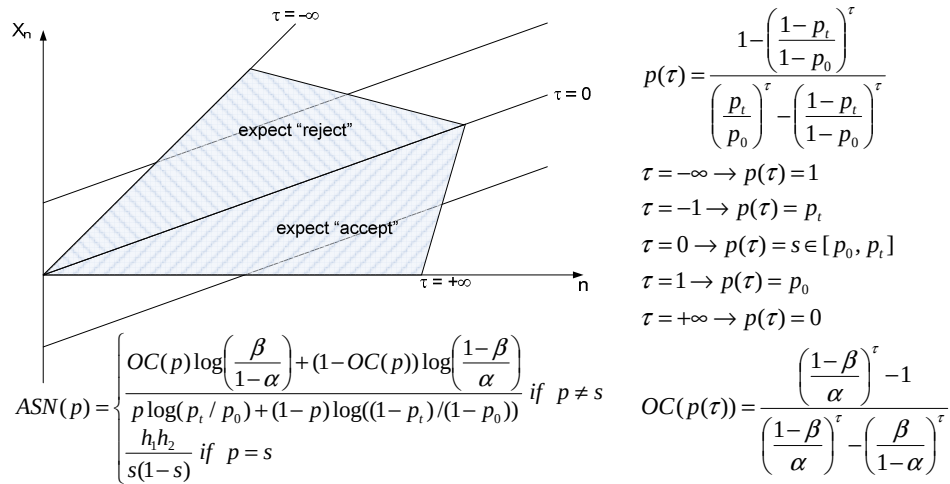
$$\text{with } h_1 = \log \left(\frac{1-\alpha}{\beta} \right) / \log \left(\frac{p_t(1-p_0)}{p_0(1-p_t)} \right) > 0$$

The example in the slide shows the procedure for : $p_0 = 0.01$, $p_t = 0.05$ and $\alpha = \beta = 0.05$. Then $h_1 = h_2 = 1.78$ and $s = 0.025$.

It can be proven that every SPRT stops **sooner or later** and that a decision is reached at that point. For practical reasons, the procedure is quit when a predefined number of inspections has been carried out. The decision is accept if at that point, the "inspection path" is closest to the accept limit, reject if otherwise.

Sequential sampling for attributes

SPRT : OC and ASN



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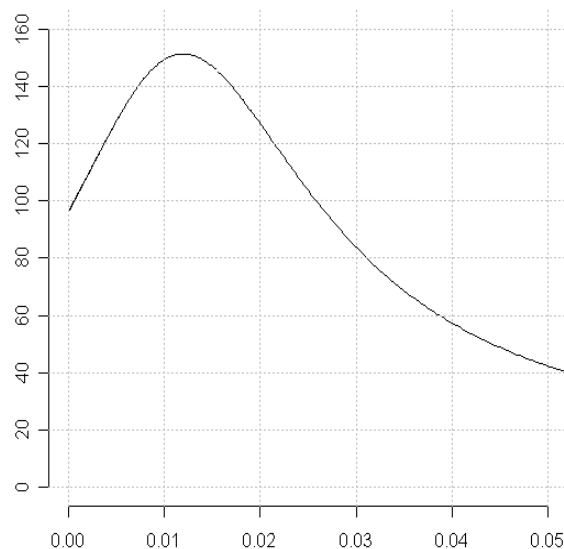
8

When $\tau = 0$ then $p = s$, which means that the average slope of the "inspection path" will be s , indeed $E(X_n) = sn$. Notice that s belongs to $[p_0, p_t]$. Positive values of τ correspond to values of p between 0 and s : in this region we expect that the procedure will stop with an "accept decision". Negative values of τ correspond to values of p between s and 1 (all items defective): in this region we expect a "reject decision". It is easily demonstrated that $OC(p(1)) = 1 - \alpha$ and $OC(p(-1)) = \beta$.

The formulas for $OC(p)$ and $ASN(p)$ are given without proof. The R-function "OC_SPRT.TXT" can be used to verify the formula for $OC(p)$ by simulation. This is left as an exercise.

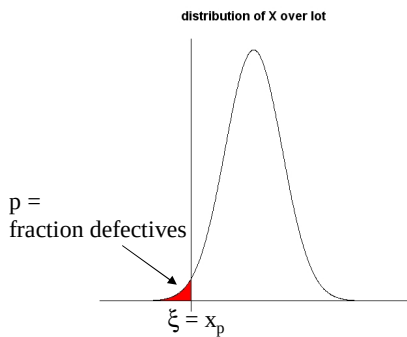
From the formula of ASN we can derive that an SPRT-procedure is indeed much more efficient than e.g. a single sample plan. When SPRT is applicable (i.e. when there are no practical obstacles), it is the recommended attributive sampling procedure. As an example we show the ASN for a sequential plan of which the OC-curve passes through (0.5 %, 95 %) and (3.5 %, 5 %). We remember that a comparable single plan, requires a sample size of 164 as was derived by the method of Peach.

ASN of SPRT



$p_0=0.005 - p_t = 3.5 - \alpha = \beta = 0.05$

Sampling plans for variables



The lot is acceptable if $p \leq AQL = p_0 \Leftrightarrow \xi \leq x_{p_0}$

Inspection procedure : draw sample of size n , calculate $\bar{X}, s$ and $Q = \frac{\bar{X} - \xi}{s}$
accept the lot if $Q \geq k \Leftrightarrow \bar{X} - ks \geq \xi$
(k is the counterpart of c for a single attributive plan)

given n , k can be determined such that $OC(AQL = p_0) = 1 - \alpha : k \cong t(1 - \alpha, p_0, n)$

$$OC(p) \cong 1 - \Phi\left(\frac{(z_p + k)\sqrt{n}}{(1 + k^2/2)^{1/2}}\right) \Rightarrow \begin{cases} k = \frac{z_{p_t} z_\alpha + z_{p_0} z_\beta}{z_{1-\alpha} + z_{1-\beta}} & \text{so that :} \\ n = \frac{(z_{1-\alpha} + z_{1-\beta})^2 (1 + k^2/2)}{(z_{p_t} - z_{p_0})^2} & \begin{aligned} OC(AQL = p_0) &= 1 - \alpha \\ OC(LQL = p_t) &= \beta \end{aligned} \end{cases}$$

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9

Let X be some quality characteristic, normally distributed over a large lot. (Note that the lot must be large if we want to consider X as having a continuous distribution.) Let ξ be a lower specification limit for the values of X , i.e. an item is considered defective if the measured X -value is below ξ .

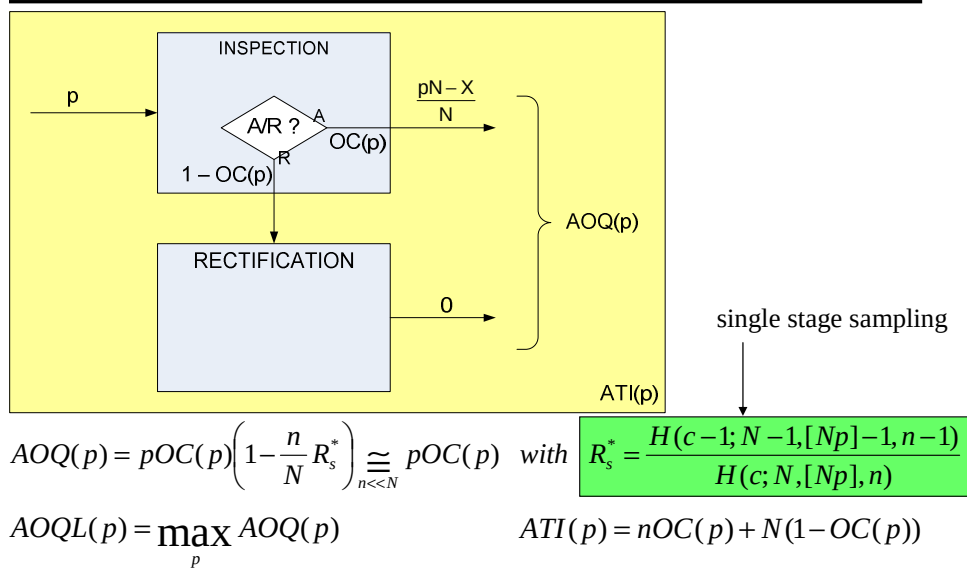
To prove that, given n , k has to be taken equal to $t(1-\alpha, p_0, n)$ to have a producer's risk of α , recall that $\bar{X} - ks$ is a $(1-\alpha, 1-\beta)$ lower tolerance limit if $k = t(\alpha, \beta, n)$, i.e. with confidence $1-\alpha$ it can be stated that a fraction of the population of at least $1-\beta$ will be larger than $\bar{X} - ks$ (see K&Z § 6.5 p. 208). This is equivalent to $X_\beta \geq \bar{X} - ks$

So by putting $k = t(1-\alpha, p_0, n)$, with probability $\alpha : X_{p_0} = \xi \geq \bar{X} - ks$ in which case the lot is rejected. The producer's risk is α , or $OC = 1 - \alpha$.

The formula for the OC-curve is given without proof. The derivation of the formulas for n and k so that the OC-curve will pass through the points $(p_0, 1-\alpha)$ and (p_t, β) is left as an exercise. As an example we calculate the variable plan, with OC-curve passing through (0.5 %, 95 %) and (3.5 %, 5 %). We find $k = 2.1939$ and $n = 64$. Compare this to a single plan for attributes. This shows the superiority of variable plans with respect to efficiency. This is a consequence of the fact that variable plans make better use of the data at hand, or in other terms : translation of measured values to attributes involves loss of information. On the other hand sampling plans for attributes have as advantages : no dependency on assumptions about the form of the distribution of the characteristic, can be used for inspections based on combinations of characteristics.

The procedure of acceptance sampling for variables can be generalized to upper and lower specification limits, to double and sequential sampling. Further information can be found in DUNCAN. The inspection procedure with a single specification limit is called a form 1 inspection. When there are 2 specification limits, the procedure is called form 2.

Rectifying inspection



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10

Normal product inspection procedures only exert indirect influence on incoming quality. Suppliers will try to deliver lots of sufficient quality that have reasonable chance to pass the inspection. Rectifying inspection is a procedure with direct impact on product quality. Lots that fail to pass the inspection stage are subjected to rectification. This is complete inspection with replacement of all defective items. Lots that pass the inspection stage are not going to rectification. It is assumed here that the tests are non-destructive, that all defective items found in the sample(s) are replaced by good ones and that the sample is replaced in the lot prior to further processing. Under these assumptions a lot with an original fraction defectives of p is either transformed into a lot with a fraction defectives of $(pN - X)/N$ (probability $OC(p)$) or into a lot with no defectives, after passing the rectification stage (probability $1 - OC(p)$).

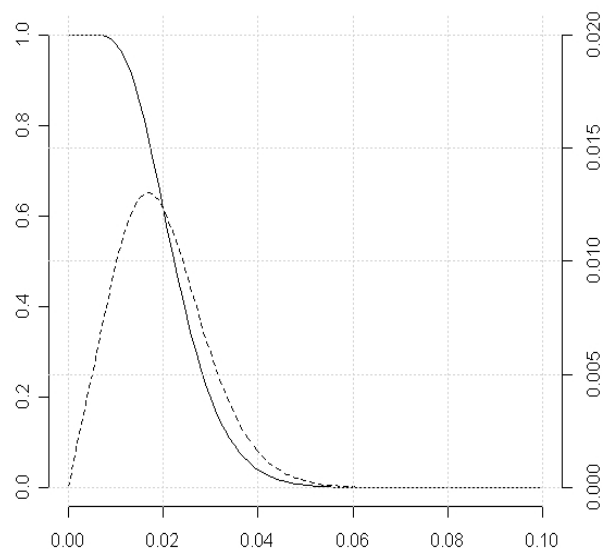
On average a stream of lots all having a fraction defectives of p is transformed into a stream with average fraction defectives $AOQ(p)$. $AOQ = \text{Average Outgoing Quality}$. The proof of the formula for $AOQ(p)$ shown on the slide follows directly from

$$AOQ(p) = \sum_{i=0}^c \left(p - \frac{i}{N}\right) h(i; N, [Np], n)$$

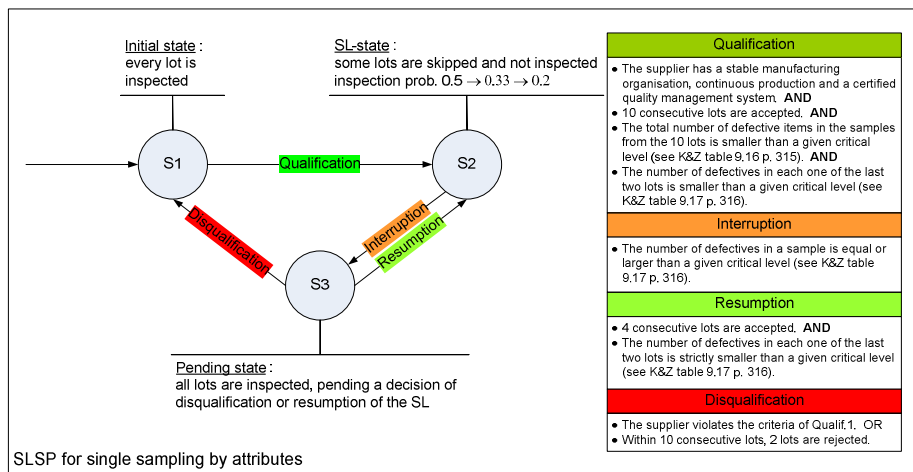
by straightforward, but rather laborious calculation. The formula for R_s^* given on the slide assumes single stage sampling. For multi stage sampling, the formulas become considerably more complex. The maximum value of the AOQ-curve is called the Average Outgoing Quality Limit (AOQL). The chart shows OC, AOQ and AOQL for a single stage inspection by attributes with $N = 1000$, $n = 250$ and $c = 5$. The AOQL-value is approximately 1.3 %, corresponding to the original percentage defectives of about 1.8 %.

ATI stands for the Average Total Inspection effort to be made. H. Dodge & H. Romig developed single and double plans for attributes with given AOQL and minimized ATI at a specified p . These plans offer a given quality protection at minimized inspection cost. Further reading, see e.g. A.J. DUNCAN "Quality Control and Industrial Statistics".

OC and AOQ for (250,5) and $N = 1000$



Skip-lot sampling plans for attributes



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11

If a supplier keeps shipping high quality lots, it makes sense to subject his lots to some form of reduced inspection, leading to savings on inspection costs. When skip-lot sampling is applied, according to prescribed probabilities, some of the lots will be skipped for inspection. At first the skipping probability is 0.5, so only half of the lots are inspected. If the inspected lots continue to be almost free of non conforming items, the skipping probability is further increased (to 0.66 and eventually to 0.8).

In a skip-lot sampling plan (SLSP), there are three states. In the initial state all lots are inspected. When quality proves to be good, the procedure goes to the skip-lot state (SL) where only a fraction of the lots are inspected. Normally the procedure stays in the SL-state, but when the inspection results deteriorate, inspection is increased again. This is the pending state. From this temporary state, the procedure either goes back to the SL-state, if inspection results improve again, or to the initial state. Both in the initial state and in the pending state, all lots are subjected to inspection, but the requirements for resumption of the SL-state are less stringent than the initial qualification requirements. The state transition rules shown on this slide are for single sampling by attributes, other rules apply for other sampling plans.

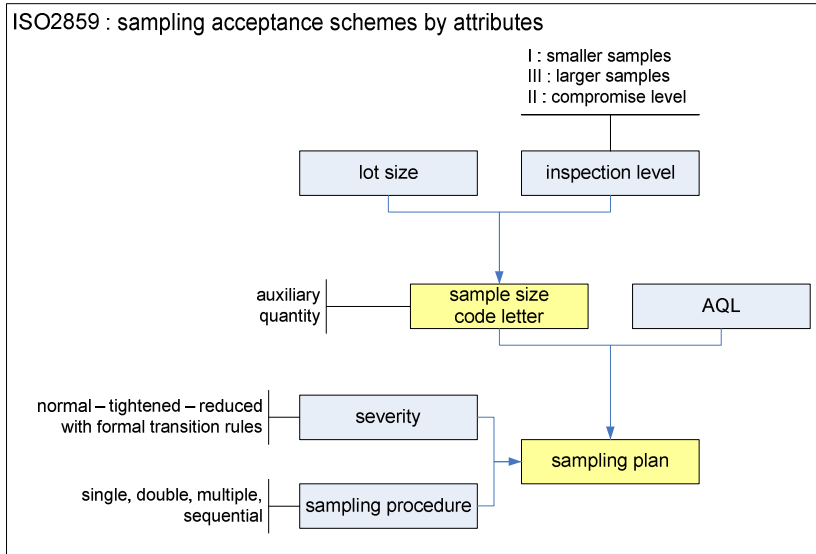
For the SLSP to be effective, the fraction of defectives level of the supplier should be smaller than half the AQL. This can be illustrated by the following example. The sampling plan (80,2) applied to lots of size 1000, has an AQL of 1 % (verify this as an exercise). Assuming that the last 10 lots were accepted, what is the probability of qualification for the SL-state ? From table 9.16 we learn that the cumulative number of defectives should be smaller than 4 (ALQ = 1 % and total number of inspected items = 800, look for the next larger value). From table 9.17 we learn that the last two lots should be free of any defectives. Combining both conditions, we conclude : the first 8 samples should have no more than 3 defectives, and the last two should be free of defectives. The probability to have this condition is (p = overall probability of producing a defective item) :

$$P_q = B(3;640,p) \times B^2(0;80,p)$$

(B = binomial c.d.f.). When $p = \text{AQL} = 1\%$, then $P_q = 2.4\%$; when $p = \text{AQL}/10 = 0.1\%$, then $P_q = 85\%$. A supplier who is producing "on the edge" (with respect to fraction defectives) will not likely qualify for skip-lot inspection. On the contrary a supplier who is delivering high quality lots, will have high probability of qualifying for the SL-state. This illustrates the importance of maintaining high-quality production processes. Being in the SL-state, all inspected lots should show no defective items or the SL-state is interrupted. Return to the SL-state is only possible when 4 consecutive lots are accepted, of which the last two are free from defectives. Note that it is possible for a supplier to have all his lots accepted, without ever reaching the SL-state.

The SLSP described on this slide is added to the ISO 2859 standard on acceptance sampling by attributes (see next slide).

International standards – sampling systems



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12

Sampling systems provide support for finding suitable inspection plans without the burden of mathematical derivations. A sampling system consists of collections of sampling plans, ordered in tables, together with meta information for navigation to the suitable inspection procedure. Some of these sampling systems have been adopted as international standards, e.g. ISO2859, sampling plans for attributes and ISO3951, the counterpart of ISO2859, for variable inspection. Both standards are general in nature and can be applied for inspection of a variety of products. Besides general purpose sampling systems, there exist domain specific systems, where the meta information for navigation relies on specific knowledge. An example are the ICMSF criteria for inspection procedures to be used in food industry. (ICMSF = International Commission on Microbiological Specifications for Foods).

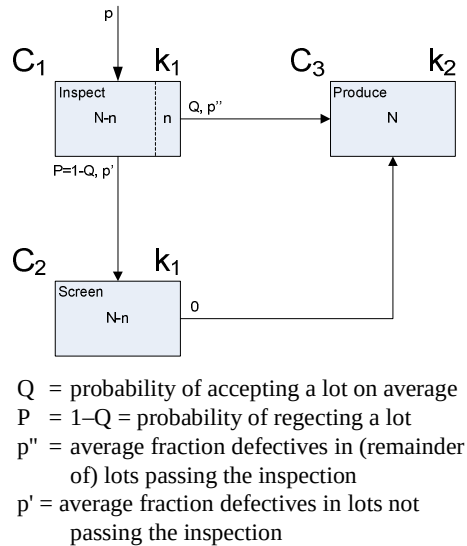
The scheme on this slide, shows the use of ISO2859. This standard goes back to the early military standard MIL-STD-105. It is designed to inspect lots from a variety of suppliers that are required to satisfy specified quality levels for the products. These quality levels are indexed by AQL. Besides AQL, there are 4 additional quantities needed for the selection of a sampling plan : lot size, inspection level, inspection severity and sampling procedure. The inspection level can take the values I, II and III (apart from the special levels S1 to S4). Level I should be taken when sampling is expensive (e.g. destructive) or laborious, III should be taken when sampling is inexpensive. II is a compromise between I and III. Level I will lead to smaller samples, level III to larger samples. Based on the lot size and the inspection level, the so-called Sample Size Code Letter (SSCL) is determined. Note that in ISO2859 larger lots will be inspected with larger samples. In ISO2859 lots can be subjected to normal, tightened and reduced inspection. Like in skip-lot sampling procedures, there are formal rules to switch between these three states (see the file "tabellen AS.doc"). The standard contains tables for various sampling procedures : from single to sequential. For each combination of severity and procedure there is a table. This tables must be entered with the AQL and the SSCL to retrieve the suitable plan.

An example : lots of size 1000 will be subjected to normal inspection at level II using a single sampling plan. The AQL is 1 %. From "tabellen AS.doc" we find : SSCL = J, plan = (80,2).

Deming's inspection criterion

⇒ Sketch of the situation :

- Parts are used in assembly ; a defective part makes the assembly fail ; this entails a unit cost k_2 (for dismantle, repair and reassemble).
- Parts are delivered in lots of size N . Lots are subjected to single stage attributive acceptance sampling (n, Ac). The unit inspection cost is k_1 . Any defective part found in inspection is replaced from a supply S by a non defective one.
- Rejected lots are screened and fully rectified.
- The part's production process is stable at level p (e.g. by using SPC). This implies that the number D of defective parts in a lot, is binomially distributed with $E(D) = Np$ and $\sigma(D) = \sqrt{Np(1-p)}$
- It can be expected that $k_1 \ll k_2$.



Deming's inspection criterion

⇒ Total quality cost = $TC = C_1 + C_2 + C_3$; this is the cost for inspection, replacing any found defectives and rework in case of assembly failure.

$$\Rightarrow C_1 = \frac{nk_1}{q}$$

$$C_2 = (N - n)(1 - Q) \frac{k_1}{q}$$

$$C_3 = (N - n)p''Qk_2$$

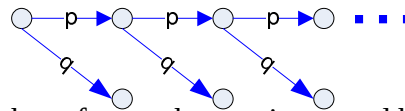
$$\Rightarrow TC = Nk_1 \left[\frac{1}{q} + Q \left(\frac{k_2}{k_1} \right) \left(1 - \frac{n}{N} \right) \left(p'' - \frac{k_1}{k_2} \frac{1}{q} \right) \right]$$

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14

Calculating C_1 :

Parts are inspected and when found defective replaced by a part from the supply S. In its turn the replacement part is inspected and when found defective, replaced, and so on (see the scheme).



The table shows the number of parts that are inspected before a non defective part is found, and the corresponding probability ($q = 1 - p$) :

<u>nb of parts inspected</u>	<u>probability</u>
1	q
2	pq
3	p^2q
...	...

Therefore the average nb. of parts inspected before replacement by a non defective part is

$$q \sum_{i=1}^{+\infty} ip^{i-1} = \frac{q}{(1-p)^2} = \frac{1}{q}$$

The associated cost is k_1/q ; since n parts are inspected, it follows that $C_1 = nk_1/q$.

Calculating C_2 :

C_2 is calculated in the same way, taking into account that when the lot is rejected, the whole remainder of the lot is inspected.

Deming's inspection criterion

- ⇒ One can prove that (see R-function DEMING.TXT)
- if $p < k_1/k_2$ then TC is minimal for $n = 0$ (no inspection)
 - if $p > k_1/k_2$ then TC is minimal for $n = N$ (screening)
 - k_1/k_2 is called the break-even quality

⇒ Conclusions :

- If the process is stable at a given percentage of defectives, then either no inspection or 100 % inspection is cost optimal
- Deming's assumption is that the production process is stable. Sampling plans do not make this assumption and are designed for catching shifts in process levels.
- Stability means that the company has absorbed the cost of SPC as an internal overhead cost or as a piece cost. Deming's assertion then is that assuming up front the cost of SPC implementation is cheaper in the long run than doing business in an environment where a process may go out of control undetected until its output undergoes acceptance sampling. Deming calls this a state of chaos, to be abandoned as soon as possible.

IR18

15

Some technical extras :

$$Q = OC(p) = \sum_{D=0}^N \pi_D \beta_D$$

π_D is the probability of having exactly D defectives in the lot = binomial = $b(D;N,p)$.

β_D is the probability of accepting such a lot = hypergeometrical = $H(Ac;N,D,n)$.

(See function OC_SAI.TXT.)

$$p'' = \sum_{D=0}^N \frac{\pi_D}{\beta_D} \sum_{j=0}^{Ac} \beta_j \frac{D-j}{N-n} > \sum_{D=0}^N \frac{\pi_D}{\beta_D} \sum_{j=0}^{Ac} \beta_j \frac{D}{N} = \sum_{D=0}^N \frac{\pi_D}{\beta_D} \frac{D}{N} \sum_{j=0}^{Ac} \beta_j = \sum_{D=0}^N \pi_D \frac{D}{N} = p$$

(See function OQ_SAI.TXT.)

Certified quality provider

- ⇒ Inspection (quality ladder, level 2) is waste (cf. lean manufacturing) as it adds no value to the product.
 - ⇒ It is better to work with certified quality providers
 - In this case it is recommended (required) that :
 - The supplier organises ongoing SPC of his production processes (quality ladder level 3).
 - The supplier's quality management system is being audited regularly (by the customer and/or by certification organisations).
 - Sporadically, lots are subjected to quality inspection, as to confirm and ensure that the quality level does not drop below the accepted standards (this is a matter of prudence).
 - This enables “ship to stock” agreements, where the supplier supplies product directly to the customer's stockroom, without any incoming inspection.
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IR18

16

It may be remarked that the organisation of SPC at the supplier's premises, also entails costs. This is of course true, but compared to pure post hoc inspection, SPC has the following fundamental advantages :

- It is organized as a natural part of the production process, therefore it may be organized in a more efficient way.
- It is preventive in nature (unless charts for non conforming product). It aims at producing without defectives.
- It forces the supplier into a system of reflecting upon the sources of variation in his production processes, and to constantly improve quality.

Building a quality management system, based on prevention is simply a more advanced approach, to be preferred over assuring quality based on mere inspection.

Exercises

⇒ construction of sampling plans for attributes

1. Construct a single stage inspection plan for attributes, giving lots with 2 % defectives an acceptance probability of 95 % and lots with 4 % defectives an acceptance probability of 5 %.
2. Construct a double sampling plan offering the same protection.
3. Compare the OC-curves of both plans.
4. What is the probability that a lot with 3 % defectives is either accepted or rejected by the double plan after the first sample ?
5. Compare both plans with respect to inspection cost. What plan is to be preferred ?

⇒ inventive inspection

A quality inspector knows that the single plans (56,0) and (95,1) have the same LQL(*)-value (check this). Normally, for reasons of inspection cost, lots are inspected with the plan (56,0). However when only 1 defective is being found, then 39 extra items are tested. When the 39 items all prove to be conforming, the lot is after all accepted, since 95 items were tested with only 1 defective. Please comment on this procedure.

(*) : LQL : percentage defectives with acceptance probability of 10 %.

Exercises

⇒ sample size vs. lot size

1. Does it seem intuitively clear that larger lots should be inspected using larger samples ? What criteria can be considered to justify this viewpoint ?
2. For a given single plan (n,c) investigate the effect of an increasing lot size N on the OC-curve of the plan.
3. In ISO2859 larger lots lead to larger samples for inspection (check this). Taking into account the answers to the above questions, comment on this procedure.